

Problem Set
Question Sheet
Quantitative Macroeconomics

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*Please submit your answers no later than **Tuesday, 19 December, 12.00 (noon)**. All computer codes (files named **Q1.m**, **Q2.mod**), figures (files named **Q1_1.pdf**, **Q1_2.pdf**, **Q2_1.pdf**, **Q2_2.pdf**), and written answers, either hand-written or typed (files named **Answers.pdf**), must be sent by email to marco.graziano@unil.ch and aurelien.eyquem@unil.ch. Files should be sent as a zipped folder named as your name(s) and surname(s) separated by a _ character.*

1 Large Crises and Expectation Formation

Consider an economy in which occasionally large crises of stochastic length occur, for example, due to a sudden pandemic world crisis. Let the indicator $s_t \in \{0, 1\}$ describe the state of the economy in period $t = 1, 2, \dots, T$. If the economy is in a normal phase at time t , then $s_t = 0$, and $s_t = 1$ if it is in a recession. Assume that $T = 50,000$.

In period $t = 1$, the economy starts in normal times, not in a crisis, i.e. $s_1 = 0$. For all periods $t > 1$, $\Pr(s_t = 0 | s_{t-1} = 0) = 0.95$ and $\Pr(s_t = 1 | s_{t-1} = 1) = 0.4$, i.e. the economy remains in normal times with probability 0.95 if it was not previously in a large recession and it stays in a recession with probability 0.4 if it was already there in the last period.

Consequently, the probabilities of state changes are $\Pr(s_t = 1 | s_{t-1} = 0) = 0.05$ and $\Pr(s_t = 0 | s_{t-1} = 1) = 0.6$.

*Write an m-file in Matlab to answer the questions below and save it as **Q1.m**.*

1. Generate a $T \times 1$ -vector u where each element is drawn from the standard uniform distribution.
2. Calculate the mean of the vector u .
3. Simulate the stochastic process $\{s_1, s_2, \dots, s_T\}$. (*Hint: You can use u_t to determine whether the state switches in t or not. For each period, write a conditional statement based on whether the value of u_t is above or below the probability for the relevant pair s_{t-1} and s_t .*)
4. Calculate the share of time periods in which the economy is in a recession. (*Hint: The number of periods in which the economy is in a recession is equal to the the mean of $\{s_1, s_2, \dots, s_T\}$.*)

Suppose now that there is a representative agent that forms expectations about the state of the economy, as it cannot fully observe it. In every period $t \in \{1, 2, \dots, T\}$, the agent's expectation formed in previous period $t-1$ is y_t , while the realized state is s_t . Expectations are equal to the actual state of the economy in the first period, so $y_1 = s_1$. Afterwards, they are formed according to the following process for $t \in [2, 10]$

$$y_t = y_{t-1} + \lambda(s_{t-1} - y_{t-1}),$$

and according to the following process for $t > 10$

$$y_t = y_{t-1} + \lambda \left(\frac{1}{10} \sum_{p=1}^{10} (s_{t-p} - y_{t-p}) \right)$$

Suppose that the realized state evolves according to the stochastic process derived above and that $\lambda = 0.25$.

*Write the code to answer the following questions into the m-file **Q1.m**.*

5. Simulate the expected state for all T periods. (*Hint: You can use “for” statements to distinguish between $t = 1, t \in [2, 10]$ and $t > 10$.*)
6. Plot $t = [400, 500]$ of both $\{s_t\}_{400}^{500}$ and $\{y_t\}_{400}^{500}$ in the same figure. Label the figure appropriately. *Save the figure as **Q1_1.pdf**.*
7. Calculate the root mean squared error of the expectation $\sqrt{\frac{1}{T} \sum_{t=1}^T (s_t - y_t)^2}$
8. Calculate the root mean squared error as above for $\lambda = 0.25, 0.3, 0.35, 0.4$ using a loop (still based on $T = 50,000$ for each of the time the model is simulated). Make a plot that has λ on the horizontal axis and the corresponding root mean squared error on the vertical axis. Label the figure appropriately. For which of the plotted values of λ are expectations most accurate? *Save the figure as **Q1_2.pdf**.*

2 Shocks and Business Cycles

There is an economy with two representative agents: a household and a firm. The profit-maximizing firm is a price taker in the markets for its inputs and sells the goods that it produces in a fully competitive market. In each period t , output Y_t depends on the prevailing level of technology A_t as well as the amount of capital K_{t-1} (installed at beginning of period t) and labor N_t used as inputs. The production technology is described by

$$Y_t = A_t K_{t-1}^\alpha N_t^{1-\alpha}$$

with $0 < \alpha < 1$.

The representative household maximizes expected utility

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \frac{1}{1-\sigma} \left(C_{t+s} - \theta_{t+s} \frac{N_{t+s}^{1+\phi}}{1+\phi} \right)^{1-\sigma}$$

subject to the budget constraint

$$C_t + K_t - (1-\delta)K_{t-1} = W_t N_t + R_{kt} K_{t-1}$$

Above, C_t is consumption, W_t is the real wage, and R_{kt} is the gross real interest rate. Moreover, $0 < \beta < 1$, $\sigma > 1$, $\phi \geq 0$, and $0 < \delta < 1$ are parameters.

The evolution of A_t and θ_t is given by

$$\begin{aligned} \log A_t &= \rho_A \log A_{t-1} + \varepsilon_t^A \\ \log \theta_t &= \rho_\theta \log \theta_{t-1} + \varepsilon_t^\theta \end{aligned}$$

with $\varepsilon_t^A \sim N(0, \sigma_A^2)$, $\varepsilon_t^\theta \sim N(0, \sigma_\theta^2)$, $0 < \rho_A < 1$, and $0 < \rho_\theta < 1$.

Employ the following parameterization in your model simulations.

$$\begin{aligned} \beta &= 0.96 \\ \sigma &= 3 \\ \phi &= 4 \\ \alpha &= 0.33 \\ \delta &= 0.075 \\ \rho_A &= 0.45 \\ \sigma_A &= 0.005 \\ \rho_\theta &= 0.6 \\ \sigma_\theta &= 0.01 \end{aligned}$$

1. Which economic interpretation can you give for θ_t ?
2. Derive the first-order conditions of the household's utility maximization problem and combine them to get a Euler equation and a labor supply condition.
3. Derive the first-order conditions of the firm's profit maximization problem.
4. Calculate the steady state value of *all* endogenous model variables using the parameter values given above. Assume that $A = 1$ and $\theta = 1$ hold in steady state.
(Recall that it can be useful to introduce auxiliary variables. Also note that household's budget constraint is part of the equilibrium.)
5. Use Dynare to calculate the impulse response functions (IRFs) of A_t , C_t , Y_t , K_t , N_t , R_{kt} and W_t to a positive realization of ε_t^A . Employ a first-order approximation of the model and graph the IRFs for 50 periods. Give a detailed economic explanation for the shape of the IRFs.
*Save the mod-file used to generate the IRFs as **Q2.mod** and save a graph of the IRFs as **Q2_1.pdf**.*
6. Calculate the IRFs of θ_t , C_t , Y_t , K_t , N_t , R_{kt} and W_t to a negative realization of ε_t^θ . Employ a first-order approximation of the model and graph the IRFs for 50 periods. Interpret the shape of the IRFs carefully again.
*Add the code used to generate the IRFs to the mod-file **Q2.mod** and save a graph of the IRFs as **Q2_2.pdf**.*
7. Compare the response of the real wage in both sets of IRFs calculated above. Does it move in the same direction? Why or why not?